Task : Malware Analysis
Intern ID:- 296

Poc:-

Malware Assigned to Analyzie:- Trojan.Agent.CMGN
117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2

Summary: The analyzed file, identified by the hash 117da274f4078bdd6778aad61d96c44100ccae59194202c166ce5f4be78b2, is a malicious executable. A total of 66 out of 72 security vendors flagged the file as malicious, resulting in a community score of -13. The file is identified as a Trojan and a ransomware dropper.

Detection Breakdown: The majority of security vendors detected the file as malicious, with many specifically identifying it as a Trojan or ransomware. The following are some key detection names and classifications from the provided screenshots:

- Ransomware:
  1. Ransom.Exroute.A4 (QuickHeal)
  2. Ransom_CERBER.SMALY0 (TrendMicro, TrendMicro-HouseCall)
  3. Ransomware-GEG/B840A2FFF66D (Trellix ENS)
  4. Win32.Ransom.Lucky (VBA32)
  5. Ransom.Win32.ccf4 (Gridinsoft)
  6. Ransom.Win32.Lucky.A (Microsoft)

- Trojans:
  1. Trojan.Kryptik!1.AD51 (Rising)
  2. Trojan.Win32.CryptLucky.EL (ViNTR)
  3. W32.Trojan.Gen (Webroot)
  4. Trojan.Win32.Lucky.670208.B (ViRobot)
  5. Trojan.Agent.CMGN (VIPRE, AllCloud, Bkav Pro, cScon, Emsisoft, MaxSecure)
  6. Generic.Trojan.Win32.907334 (Zillya)
  7. Packed.Generic.493 (Symantec)
  8. Trojan.Generic.Win32.539879 (Zoner)
  9. Trojan.Win32.Recoryc.toxv (Lionic)

- Other Malicious Classifications:

1. Malicious (SecureAge, DeepInstinct, Threatgrid)
2. Static-AI - Malicious PE (SentinelOne)
3. Mal/Elenoocka-E (Sophos, ZoneAlarm)
4. Generic.Malware (TEHTRIS)
5. Malicious.high.ml.score (Trapmine)
6. Trojan.Win32.Recoryc (Antiy-AVL)
7. Malicious (High Confidence) (Elastic)
8. Heuristic.HEUR/AGEN:1315815 (WithSecure)
9. Heuristic.Win32.Generic (K7AntiVirus)
10. Trojan-Downloader.Win32.WannaCry.d (Malwarebytes)

VirusTotal - File - 117da274f4076

https://www.virustotal.com/gui/file/117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2/detection

117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2

Sign in | Sign up

66 / 72

Community Score -13

66/72 security vendors flagged this file as malicious

Reanalyze | Similar | More

117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2

malware.exe

peexe | spreader | runtime-modules | direct-cpu-clock-access | detect-debug-environment | long-sleeps

Size 654.50 KB

Last Analysis Date 6 days ago

EXE

DETECTION | DETAILS | RELATIONS | BEHAVIOR | COMMUNITY 13

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan.locky/cmgn

Threat categories trojan | ransomware

Family labels locky | cmgn | reconyc

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Win-Trojan/Lukitus3.Exp	Alibaba	Trojan:Win32/Locky.9b2d8e81
AliCloud	Ransomware:Win/Locky.L	ALYac	Trojan.Agent.CMGN
Antiy-AVL	Trojan/Win32.Reconyc	Arcabit	Trojan.Agent.CMGN
Arctic Wolf	Unsafe	Avast	Win32:Malware-gen
AVG	Win32:Malware-gen	Avira (no cloud)	HEUR/AGEN.1315815

VirusTotal - File - 117da274f4076

https://www.virustotal.com/gui/file/117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2/detection

117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2

Sign in | Sign up

BitDefender	Trojan.Agent.CMGN	Bkav Pro	W32.AIDetectMalware
ClamAV	Win.Ransomware.Locky-6336169-0	CrowdStrike Falcon	Win/malicious_confidence_100% (W)
CTX	Exe.trojan.locky	Cynet	Malicious (score: 99)
DeepInstinct	MALICIOUS	DrWeb	Trojan.Encoder.13570
Elastic	Malicious (high Confidence)	Emsisoft	Trojan.Agent.CMGN (B)
eScan	Trojan.Agent.CMGN	ESET-NOD32	Win32/Filecoder.Locky.L
Fortinet	W32/Kryptik.FVZV!tr	GData	Win32.Trojan.Kryptik.IT
Google	Detected	Gridinsoft (no cloud)	Ransom.Win32.Locky.ccls4
Huorong	HVM:Trojan/MalBehav.gen!E	Ikarus	Trojan.Win32.Tofsee
Jiangmin	Trojan.Reconyc.hcv	K7AntiVirus	Trojan (0051918c1)
K7GW	Trojan (00515e9f1)	Kaspersky	HEUR:Trojan.Win32.Generic
Lionic	Trojan.Win32.Reconyc.toVV	Malwarebytes	Malware.AI.1687953590
MaxSecure	Trojan.Malware.11316871.susgen	McAfee Scanner	Real Protect-LS!0B40A2FFF66D
Microsoft	Ransom:Win32/Locky.A	NANO-Antivirus	Trojan.Win32.Encoder.esjllr
Palo Alto Networks	Generic.ml	Panda	Trj/Genetic.gen
QuickHeal	Ransom.Exxroute.A4	Rising	Trojan.Kryptik!1.AD51 (CLASSIC)
Sangfor Engine Zero	Suspicious.Win32.Save.a	SecureAge	Malicious
SentinelOne (Static ML)	Static AI - Malicious PE	Skyhigh (SWG)	BehavesLike.Win32.PWSZbot.jc
Sophos	Mal/Elenoocka-E	Symantec	Packed.Generic.493
TEHTRIS	Generic.Malware	Tencent	Malware.Win32.Gencirc.10bfac8b
Trapmine	Malicious.high.ml.score	Trellix ENS	Ransomware-GEG!0B40A2FFF66D
TrendMicro	Ransom_CERBER.SMALY0	TrendMicro-HouseCall	Ransom_CERBER.SMALY0
Varist	W32/Locky.BZ.gen!Eldorado	VBA32	Trojan-Ransom.Locky
VIPRE	Trojan.Agent.CMGN	VirIT	Trojan.Win32.CryptLocky.EL
ViRobot	Trojan.Win32.Locky.670208.B	Webroot	W32.Trojan.Gen
WithSecure	Heuristic.HEUR/AGEN.1315815	Xcitium	TrojWare.Win32.Ransom.Locky.AD@77y...
Yandex	Trojan.GenAsa!7uEZyprOBpQ	Zillya	Trojan.Generic.Win32.907334
ZoneAlarm by Check Point	Mal/Elenoocka-E	Zoner	Trojan.Win32.59879
Acronis (Static ML)	Undetected	Baidu	Undetected
CMC	Undetected	Kingsoft	Undetected

Details:

File Type: PE32 executable (GUI) Intel 80386, for MS Windows

TrID:
Win32 Dynamic Link Library (generic) (25%)
Win16 NE executable (generic) (19.1%)
Win32 Executable (generic) (17.1%)
Win16/32 Executable Delphi generic (7.8%)

Creation Time: 2015-03-24 11:48:16 UTC

Compiler: The compilation timestamp of 2015-03-24 11:48:16 UTC and the linker (Microsoft Linker (14.0)) suggest the file was compiled around that time.

Hashes:
MD5: 0b40a2fff66d3c7f728b2d0e9ae861a6
SHA-1: 91f8c4b07f9b2bad94e16d6c35a5639adf6979
SHA-256: 117da274f4078bdd6778aad61d96c44100ccae59194202c166ce5f4be78b2

Names:
malware.exe
_117da274f4078bdd6778aad61d96c44100ccae59194202c166ce5f4be78b2.exe.infected
_117da274f4078bdd6778aad61d96c44100ccae59194202c166ce5f4be78b2.exe
0b40a2fff66d3c7f728b2d0e9ae861a6

PE Sections: The file contains four sections, which are typical for an executable. The high entropy values for the .data and .rsrc sections (7.7 and 7.94 respectively) are a point of concern, as high entropy can be an indicator of packed or encrypted data. This aligns with the behavioral analysis that suggests anti-analysis techniques are in use.
.text: Virtual Address 4096, Virtual Size 59680, Raw Size 59904, Entropy 4.94
.rdata: Virtual Address 65536, Virtual Size 6340, Raw Size 6656, Entropy 2.4
.data: Virtual Address 73728, Virtual Size 34807, Raw Size 34816, Entropy 7.7
.rsrc: Virtual Address 110592, Virtual Size 567584, Raw Size 567808, Entropy 7.94

Names

malware.exe
_117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2.exe.infected
_117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2.exe
0b40a2fff66d3c7f728b2d0e9ae861a6
VirusShare_0b40a2fff66d3c7f728b2d0e9ae861a6
output.111976607.txt
117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b20b40a2fff66d3c7f728b2d0e9ae861a6.CTI
output.111985901.txt
8927f611-9011-11e7-8d7a-80e65024849a.file
59e62ca1-9018-11e7-bd91-80e65024849a.file

Portable Executable Info

Header

Target Machine	Intel 386 or later processors and compatible processors
Compilation Timestamp	2015-03-24 11:48:16 UTC
Entry Point	39688
Contained Sections	4

Sections

Name	Virtual Address	Virtual Size	Raw Size	Entropy	MD5	Chi2
.text	4096	59680	59904	4.94	85bf080a60028b4d9c2ac8af3fae592b	1954345
.rdata	65536	6340	6656	2.4	2109196f96d9796236b2db3ff56d26a1	907101.88
.data	73728	34807	34816	7.7	0f45b8bf2456c40c0d8847ab72e9f0fa	55390.5
.rsrc	110592	567584	567808	7.94	1efbeda5f71dbbd7c0fe79e78175fa65	70674.17

Relations: The malware exhibits a complex network of relations, indicating its multi-stage and networked nature.

Dropped Files: The malware drops a significant number of files (40 total), including what appears to be a PGP Security Key (PWCCANLGRZ.docx). The majority are generic files, likely components of the malware or its payload. This behavior is typical of droppers, which deploy additional malicious files to the system.

Contacted Domains: The malware contacts at least four domains, all registered by MarkMonitor Inc. The domains config-edge-skype.ln-0007.ln-msedge.net and ln-0007.ln-msedge.net were recently created, which is suspicious. res.public.onecdn.static.microsoft and www.microsoft.com are likely used for masquerading or C2 communication.

Contacted IP Addresses: The malware contacts at least seven IP addresses, located in the US and Canada. This suggests command-and-control (C2) infrastructure used for communication, data exfiltration, or receiving further instructions.

Bundled Files: The malware contains six bundled files, which are likely embedded components or payloads that are unpacked during execution.

PE Resource Children: It has one PE resource child, which could be another executable or a configuration file embedded within the main file's resource section.

Graph Summary: The visual summary shows a strong relational network, highlighting the malware's reliance on multiple components, C2 communication, and the dropping of a large number of files.

Contacted Domains (4)

Domain	Detections	Created	Registrar
config-edge-skype.ln-0007.ln-msedge.net	0 / 94	2023-04-20	MarkMonitor Inc.
ln-0007.ln-msedge.net	0 / 94	2023-04-20	MarkMonitor Inc.
res.public.onecdn.static.microsoft	0 / 94	2023-05-05	MarkMonitor Inc.
www.microsoft.com	0 / 94	1991-05-02	MarkMonitor Inc.

Contacted IP addresses (7)

IP	Detections	Autonomous System	Country
150.171.22.17	0 / 94	8075	US
184.27.218.92	1 / 94	16625	US
192.168.0.33	0 / 94	-	-
20.69.140.28	0 / 94	8075	US
23.46.228.41	0 / 94	20940	US
23.46.228.49	0 / 94	20940	US
23.49.140.110	0 / 94	16625	CA

Bundled Files (6)

Scanned	Detections	File type	Name
2017-10-21	0 / 60	?	5
?	?	file	444650375f7642ba00b11ce47d998da38e1c82c1825b4a7456191d7e281ef691
?	?	file	4247c0c1a4ed298592761489b5119b608d27c32be7934e2b2cddd26d23bc35ba
?	?	file	2832abe24e5ca959ab92f9006a83574dcdb9c4497c978b46da829bf37126304c
?	?	file	d3f9a91fd709375fdd7bf5ce457a244a93f74d9f0e1d2414e001d264d368ab48

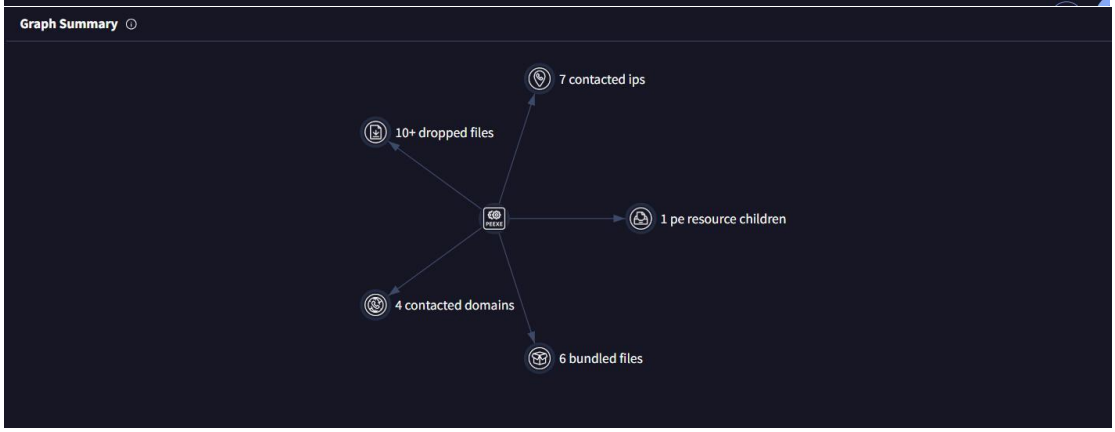
Dropped Files (40)

Scanned	Detections	File type	Name
2025-07-29	0 / 62	PGP Security Key	PWCCAWLGRE.docx
?	?	file	05f7ad0bb8cdf38378b20d8c80fb803bdd7bdc07064c06e2df9c64137bf2f606
?	?	file	15901eea0b08e9c7ea7a71cdcde3356f8eab99202690764645 3eda37ee127f5c
?	?	file	16a19763520bc5acfa82af2e5e88c3392483d7fc79726e0f774fb6b737a4042f
?	?	file	20b7b00a26f5bd9dfe18dbf35119a3397f74cb0973c69ef8174e22fcfdcccef7
?	?	file	26fb96119fca9e35aa46db174f505008c6c5c92259098a2f0fd566011a404528
?	?	file	357241ed7fd3a7870f7c06852dfeb498fe04f5912165f253118b075c94a7db02
?	?	file	36736b863d7d752e577c35fa8830eccf1e63fc5291e7d8d048ed14c1c1a135a5
?	?	file	4d257cff218a275ea687633b17375e4c10be171fe8b6e1363ad68983b7488f31
?	?	file	4d92003bb2600c8d93ea5e1dffe439ef84fc15e81c01b8503ec051221ff1b937

PE Resource Children (1)

Scanned	Detections	File type	Name
2017-10-21	0 / 60	?	5

Graph Summary



Behavioral Analysis: The dynamic analysis of the malware reveals several key behaviors:

Behavior Tags: The file is tagged with various suspicious behaviors, including crypto (indicating encryption activity), detect-debug-environment, direct-cpu-clock-access, long-sleeps, obfuscated, runtime-modules, and self-delete. These tags strongly suggest the malware is designed to evade analysis and detection.

Dynamic Analysis Sandbox Detections:
Tencent HABO: Flags the file as MALWARE EVADER RANSOM. This confirms the malware's dual functionality of evading detection and acting as ransomware.
Zenbox: Flags the file as MALWARE RANSOM, corroborating the ransomware classification.

Activity Summary:
Detections: Confirms 3 detections: 2 as malware evader and 2 as ransomware.
Network Comms: Confirms network communication over DNS and SIP protocols.
Dropped Files: Notes 11 "other" dropped files, in addition to the ones identified in the Relations section.

MITRE ATT&CK Tactics and Techniques: The malware's behavior aligns with several MITRE ATT&CK tactics, indicating its malicious purpose:

1. Execution (TA0002): The malware's primary action of executing its code on the system.
2. Persistence (TA0003): The CurrentVersion Autorun Keys Modification signature suggests it modifies autorun registry keys to ensure it runs after system reboot.
3. Defense Evasion (TA0005): Techniques like detecting debug environments, long sleeps, and self-deletion are classic defense evasion tactics.
4. Command and Control (TA0011): The contacted domains and IP addresses confirm C2 communication.
5. Impact (TA0040): The primary impact is ransomware, which involves data encryption. The "Shadow Copies Deletion" Sigma rule directly points to a common ransomware behavior to prevent data recovery.

Malware Behavior Catalog Tree:

1. Anti-Behavioral Analysis (OB0001): Confirms the use of anti-debugging and sandbox detection techniques.
2. Anti-Static Analysis (OB0002): The high entropy of PE sections and the obfuscated tag suggest techniques to prevent static analysis.

Sigma Rules: Several high- and medium-severity Sigma rules match the malware's behavior, providing specific details on its actions:

1. Shadow Copies Deletion Using Operating Systems Utilities: A critical ransomware behavior to prevent restoration of encrypted files.
2. Suspicious Volume Shadow Copy Vssapi.dll Load: Indicates the malware is loading a specific DLL to interact with Volume Shadow Copy services.
3. CurrentVersion Autorun Keys Modification: A medium-severity rule indicating a persistence mechanism.
4. File Deletion Via Del: Suggests the malware attempts to delete its own files or other artifacts to cover its tracks (self-delete).

Display grouped sandbox reports

CAPA	0	0	0	0	0	0	CAPE Sandbox	0	8	0	3	35	0
Microsoft Sysinternals	0	0	0	2	72	8	Rising MOVES	0	0	0	0	0	0
Tencent HABO	3	0	0	0	0	0	VirusTotal Cuckoofork	0	0	0	0	0	0
VirusTotal Jujubox	0	0	0	0	0	0	Zenbox	2	6	0	3	11	2

Activity Summary

Download Artifacts Full Reports Help

3 Detections	Mitre Signatures	IDS Rules	Sigma Rules	Dropped Files	Network comms
2 MALWARE 1 EVADER 2 RANSOM	9 MEDIUM 10 LOW 12 INFO	NOT FOUND	2 HIGH 2 MEDIUM 1 LOW	117 OTHER	4 DNS 6 IP

Behavior Tags

Behavior Tags

crypto | detect-debug-environment | direct-cpu-clock-access | long-sleeps | obfuscated | runtime-modules | self-delete

Dynamic Analysis Sandbox Detections

⚠ The sandbox Tencent HABO flags this file as: MALWARE EVADER RANSOM

⚠ The sandbox Zenbox flags this file as: MALWARE RANSOM

MITRE ATT&CK Tactics and Techniques

- + Execution TA0002
- + Persistence TA0003
- + Privilege Escalation TA0004
- + Defense Evasion TA0005
- + Credential Access TA0006
- + Discovery TA0007
- + Collection TA0009
- + Command and Control TA0011
- + Impact TA0040

Malware Behavior Catalog Tree

- + Anti-Behavioral Analysis OB0001
- + Anti-Static Analysis OB0002

CRITICAL 0 HIGH 2 MEDIUM 2 LOW 1

Matches rule Shadow Copies Deletion Using Operating Systems Utilities by Florian Roth (Nextron Systems), Michael Haag, Teymur Kheirkhabarov, Daniil Yugoslavskiy, oscd.community, Andreas Hunkeler (@Karneades) at Sigma Integrated Rule Set (GitHub)
↳ *Shadow Copies deletion using operating systems utilities*

Matches rule Suspicious Volume Shadow Copy Vssapi.dll Load by frack113 at Sigma Integrated Rule Set (GitHub)
↳ *Detects the image load of VSS DLL by uncommon executables*

Matches rule CurrentVersion Autorun Keys Modification by Victor Sergeev, Daniil Yugoslavskiy, Gleb Sukhodolskiy, Timur Zinniatullin, oscd.community, Tim Shelton, frack113 (split) at Sigma Integrated Rule Set (GitHub)
↳ *Detects modification of autostart extensibility point (ASEP) in registry.*

Matches rule Potentially Suspicious Volume Shadow Copy Vsstrace.dll Load by frack113 at Sigma Integrated Rule Set (GitHub)
↳ *Detects the image load of VSS DLL by uncommon executables*

Matches rule File Deletion Via Del by frack113 at Sigma Integrated Rule Set (GitHub)
↳ *Detects execution of the builtin "del","erase" commands in order to delete files. Adversaries may delete files left behind by the actions of their intrusion activity. Malware, tools, or other non-native files dropped or created on a system by an adversary may leave traces to indicate to what was done within a network and how. Removal of these files can occur during an intrusion, or as part of a post-intrusion process to minimize the adversary's footprint.*

See all

Hybrid Analysis Results:

The file was submitted to Hybrid Analysis on two separate occasions:

September 1st, 2017:
1. Threat Level: Malicious
2. Threat Score: 97/100
3. AV Detection: 97% of antivirus engines detected it.
4. AV Label: Trojan.Generic
5. Matched Indicators: 85
6. Environment: Windows 7 / 32 bit

June 29th, 2020:
1. Threat Level: Malicious
2. AV Detection: 97% of antivirus engines detected it.
3. AV Label: Trojan.Generic
4. Environment: win81

Analysis Overview:
1. Submission Name: 655KiB
2. Type: pexe (Portable Executable)
3. Mime: application/x-dosexec
4. Anti-Virus Results:
   a) CrowdStrike Falcon: Malicious (100%)
   b) MetaDefender: Malicious (26/26)

HYBRID ANALYSIS | Sandbox | Quick Scans | File Collections | Resources | Request Info | IP, Domain, Hash... | More

Search results for *117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2*

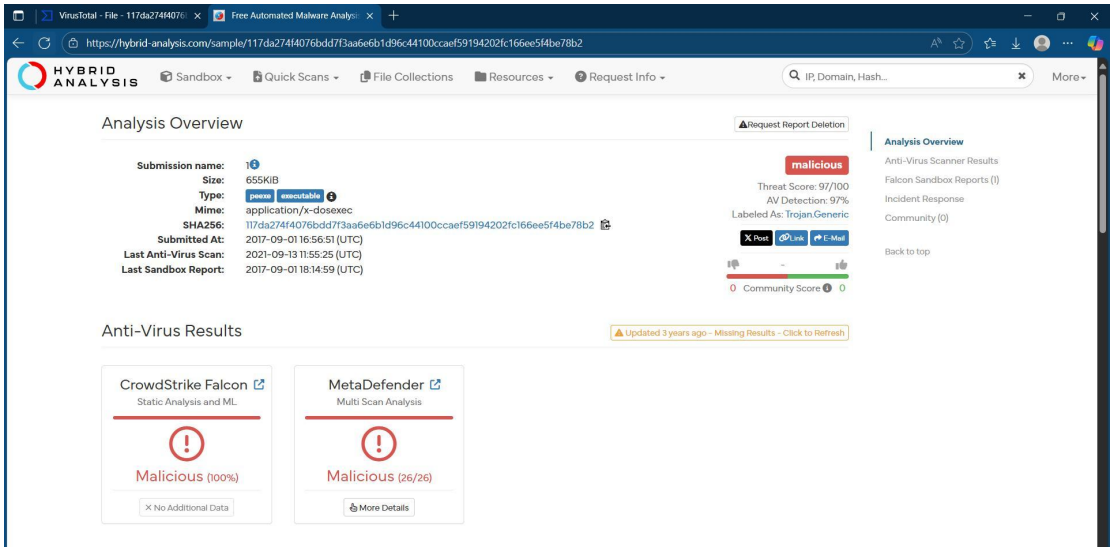
Multi-Process | Extracted Files | Sample not shared | Network Traffic | TOR analysis | Decrypted SSL traffic

Login to Download all DNS Requests (CSV) | Login to Download all Contacted Hosts (CSV) | Copy hashes | Select all

Timestamp	Input	Threat level	Analysis Summary	Countries	Environment	Action
June 29th 2020 16:45:27 (UTC)	file PE32 executable (GUI) Intel 80386, for MS Windows 117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2	malicious	AV Detection: **97%** Trojan.Generic	-	quickscan	☐
September 1st 2017 17:00:35 (UTC)	1 PE32 executable (GUI) Intel 80386, for MS Windows 117da274f4076bdd7f3aa6e6b1d96c44100ccaef59194202fc166ee5f4be78b2	malicious	Threat Score: 97/100 AV Detection: **97%** Trojan.Generic Matched **35** Indicators Show Similar Samples		Windows 7 32 bit	☐

Copy hashes | Select all

« 1 »



The consistent high threat score and detection rates across multiple submissions over several years (2017 to 2020) and on different analysis platforms (VirusTotal and Hybrid Analysis) underscore its persistent malicious nature and the stability of its detection signatures.

Conclusion: The file 117da274f4078bdd6778aad61d96c44100ccae59194202c166ce5f4be78b2 is definitively malicious. Its primary functions appear to be related to Trojan activity and ransomware deployment, with several vendors identifying it as a specific Lucky ransomware variant. The behavioral indicators, including the high entropy of certain PE sections and anti-analysis checks, suggest it employs sophisticated techniques to hinder detection and reverse engineering efforts. The extensive network of relations—dropped files, contacted domains, and IP addresses—confirms its multi-stage and networked nature, likely for C2 communication and payload delivery. The dynamic analysis confirms ransomware activity, specifically the deletion of shadow copies, a hallmark of this type of malware. The consistent results from multiple analysis platforms over time further solidify its malicious classification. This file should be considered a significant threat and immediately quarantined or removed from any system.